Department of Software Engineering
Faculty of Science & Information Technology

UniExpert

AI-Powered Expert Discovery and Academic Collaboration Platform

Capstone Project Proposal (Minimal Edition)

Course: SE 331 — Software Engineering Design Capstone Project

Semester: Fall 2026 · **Duration:** October – December 2026

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1 Executive Summary and Problem Formulation

Undergraduate and graduate students conducting capstone projects, theses, and interdisciplinary research frequently face severe friction in identifying, evaluating, and initiating collaboration with domain-specific experts, academic researchers, and industry mentors. Traditional institutional directories, keyword-based search bars, and disconnected academic websites fail to bridge this divide due to four structural deficiencies:

(1) Vocabulary Mismatch: Students express problems in plain language but lack formal knowledge of specialized academic taxonomies (e.g., searching "poultry disease detection" instead of "computational epidemiology and livestock computer vision").
(2) Institutional Silos & Cold Outreach: Academic profiles are fragmented across departmental pages, personal websites, and Google Scholar, forcing students into speculative cold emails with high rejection rates.
(3) Credential Verification Void: Self-reported claims on general professional networks cannot verify peer-reviewed contributions, departmental appointments, or active availability.
(4) Multidisciplinary Team Bottlenecks: Complex projects demand complementary teams rather than a single advisor, yet students have no automated means of discovering skill gaps in their advisory panels.

To resolve these challenges, **UniExpert** introduces an AI-powered expert discovery and academic collaboration platform operating on a *problem-first paradigm*: students articulate requirements in natural language, and the system extracts domain entities, executes dense vector similarity matching in Qdrant, computes multi-criteria hybrid ranking, and produces transparent Explainable AI (XAI) justifications.

2 Proposed Solution and Core Capabilities

UniExpert bridges students and verified mentors through five core engineering capabilities:

- **Natural Language Ingestion & Zero-Shot Skill Extraction:** Ingests unstructured problem statements and extracts primary/secondary domains and prerequisite skills using transformer-based NER.
- **Multi-Tier Expert Verification:** Implements a 4-tier badging system: Tier 1 (Institutional Email), Tier 2 (ORCID / Google Scholar Bibliometrics), Tier 3 (Departmental Appointment Audit), Tier 4 (Industry Credentials).
- **Hybrid Multi-Criteria Matchmaker:** Blends dense semantic vector retrieval with taxonomy overlap, credential authority, and real-time calendar availability.
- **Explainable AI (XAI) Justifications:** Synthesizes human-interpretable rationale for why each candidate is recommended (e.g., matching skills, relevant publications, availability status).
- **Multidisciplinary Team Builder & Skill Gap Optimizer:** Decomposes interdisciplinary projects and uses set-cover optimization to detect uncovered technical skills.

3 Mathematical Formulations

3.1 Hybrid Multi-Criteria Matching Function

The composite compatibility score $\mathcal{S}(u, e) \in [0, 1]$ between student project u and candidate expert e is formulated as a normalized linear scalarization:

$$\mathcal{S}(u, e) = w_1 \cdot \mathcal{D}(u, e) + w_2 \cdot \mathcal{K}(u, e) + w_3 \cdot \mathcal{R}(u, e) + w_4 \cdot \mathcal{V}(e) + w_5 \cdot \mathcal{A}(u, e) \tag{1}$$

where $\sum_{i=1}^{5} w_i = 1.0$ (baseline configuration: $w_1 = 0.30, w_2 = 0.25, w_3 = 0.20, w_4 = 0.15, w_5 = 0.10$).

- $\mathcal{D}(u, e) \in [0, 1]$: Domain relevance score based on hierarchical taxonomy tree distance.

- $\mathcal{K}(u,e) = \frac{|\mathcal{K}_u \cap \mathcal{K}_e|}{|\mathcal{K}_u|}$: Jaccard skill overlap between project prerequisites $\mathcal{K}_u$ and expert competencies $\mathcal{K}_e$.
- $\mathcal{R}(u,e) = \frac{\vec{v}_u \cdot \vec{v}_e}{\|\vec{v}_u\|\|\vec{v}_e\|}$: Cosine similarity between 384-dimensional dense semantic embeddings (`all-MiniLM-L6-v2`).
- $\mathcal{V}(e) \in \{0.25, 0.50, 0.75, 1.0\}$: Credential authority weight corresponding to verification Tiers 1 through 4.
- $\mathcal{A}(u,e) \in [0,1]$: Availability factor (1.0 if available, 0.5 if limited, 0.0 if in calendar blackout).

3.2 Team Skill Gap Optimization

When assembling an advisory team $\mathcal{M} = \{e_1, e_2, \ldots, e_m\}$, the collective unresolved skill gap $\Delta\mathcal{K}$ is defined as:

$$\Delta\mathcal{K} = \mathcal{K}_u \setminus \bigcup_{e \in \mathcal{M}} \mathcal{K}_e \tag{2}$$

If $|\Delta\mathcal{K}| > 0$, the platform executes targeted sub-queries to recommend complementary experts that minimize $|\Delta\mathcal{K}|$.

4 System Architecture and Technical Stack

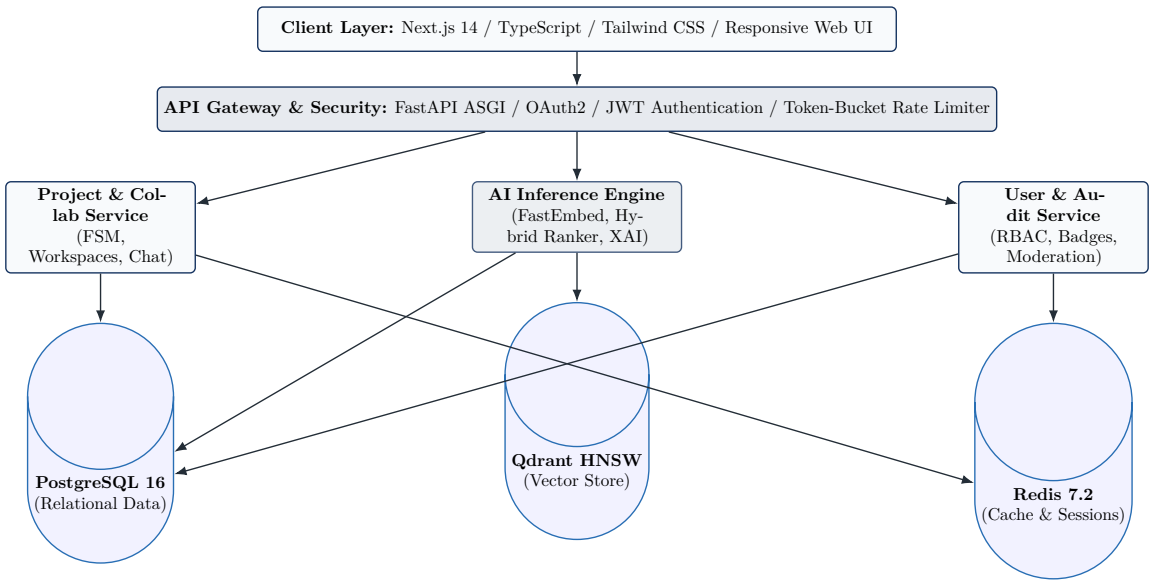


Figure 1: UniExpert High-Level Architectural Topology

Table 1: Selected Technology Stack and Justification

Layer	Technology	Architectural Rationale
Frontend Client	Next.js 14 / React	Server-side rendering, fast First Contentful Paint ($<$ 1s), responsive UI.
Backend Service	FastAPI (Python 3.11+)	High-throughput asynchronous ASGI web server; native Python ML integration.
Relational Storage	PostgreSQL 16	ACID compliance, robust foreign key constraints, JSONB support for schemas.
Vector Database	Qdrant 1.8+	High-performance Approximate Nearest Neighbor (ANN) search using HNSW indexing.
Embedding Model	`all-MiniLM-L6-v2`	384-dimensional dense semantic vectors with sub-millisecond CPU inference latency.
In-Memory Store	Redis 7.2	JWT session management, token blocklists, and rate-limiting buckets (60 req/min).

5 Implementation Scope and MoSCoW Prioritization

To guarantee successful delivery within the Fall 2026 12-week semester (October – December 2026), project features are triaged using the MoSCoW framework in Table 2.

Table 2: MoSCoW Feature Prioritization for Fall 2026 Capstone

Priority	Feature Scope	Target Delivery
Must-Have (Core MVP)	• Role-based authentication (Student, Expert, Admin) with email verification. • Multi-tier expert credential verification (institutional badge Tiers 1–4). • Natural language project statement submission and NLP skill extractor. • Qdrant dense vector search with multi-criteria hybrid ranker (Eq. 1). • Explainable AI (XAI) natural language justification generation. • Collaboration request lifecycle Finite State Machine (FSM) and workspace.	**Oct – Nov 2026** (Weeks 1–8)
Should-Have (Extended Scope)	• Multidisciplinary problem decomposition and team builder (Eq. 2). • Side-by-side candidate comparison matrix. • Automated public Google Scholar / ORCID bibliometric synchronization. • Post-collaboration double-blind mutual peer reviews.	**December 2026** (Weeks 9–12)
Could-Have (Future Work)	Real-time WebRTC audio/video rooms, cross-institutional federation, mobile apps.	Post-Capstone

6 Semester Development Plan and Milestones (Fall 2026)

The 12-week execution schedule spans October through December 2026 across three defined monthly milestones, as detailed in Table 3.

Table 3: Fall 2026 Capstone Execution Timeline and Milestone Deliverables

Phase	Weeks	Key Deliverables & Activities	Milestone
Month 1 (October 2026)	Weeks 1–4	• Requirements elicitation and formal IEEE 830 SRS documentation. • Literature review on expert finding systems and vector search. • Database schema design (PostgreSQL), API contracts, and CI/CD setup.	**M1: System Blueprint** (Requirements & Design)
Month 2 (November 2026)	Weeks 5–8	• Implement OAuth2/JWT authentication and profile verification services. • Build NLP skill extraction pipeline and integrate Qdrant vector database. • Deploy hybrid multi-criteria ranking engine and collaboration FSM.	**M2: Core Platform** (Functional Matchmaker)
Month 3 (December 2026)	Weeks 9–12	• Multidisciplinary team builder and skill gap optimizer. • Empirical IR benchmarking (BM25 vs. Vector vs. Hybrid: Precision@K, NDCG@K). • User Acceptance Testing (UAT), comprehensive report, and final defense.	**M3: Final Defense** (Evaluation & Sign-Off)

7 Risk Management and Expected Capstone Outcomes

Table 4: Project Risk Matrix and Engineered Mitigations

Identified Risk	Likelihood	Impact	Engineered Mitigation Protocol
Profile Cold-Start	High	High	Pre-seed database with public faculty profiles; enable automated ORCID sync.
Vector Search Latency	Low	High	Pre-compute embeddings; utilize HNSW indexing; cache top domain queries in Redis.
Credential Falsification	Med	High	Enforce `.edu` email verification; require admin review for Tier 3 faculty badges.
Faculty Availability Fatigue	High	Med	Monthly request quotas (max 5 pending); calendar blackout toggles.

7.1 Expected Capstone Outcomes

Upon completion in December 2026, UniExpert will deliver:

1. A production-grade, accessible web platform conforming to modern software engineering standards.
2. An empirical benchmarking evaluation proving statistically significant improvements in recommendation accuracy ($NDCG$@5, Precision@5) over classical keyword and pure vector baselines.
3. A formal capstone thesis documentation package prepared for university repository archiving.